

You must show your work to receive credit. Unless stated otherwise, all problems are worth 5 points.

1. Evaluate the following [4 pts each]:

a. -2^{-3}

b. $\log_{\pi} 1$

c. $\ln(e^3)$

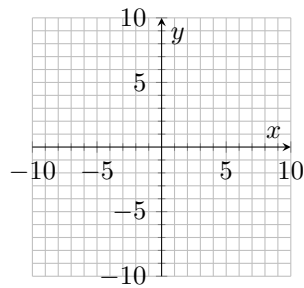
d. $\lim_{x \rightarrow -\infty} 7^x$

e. $\lim_{x \rightarrow \infty} \log_7 x$

2. Let $f(x) = 2^x - 3$.

a. Graph $f(x)$.

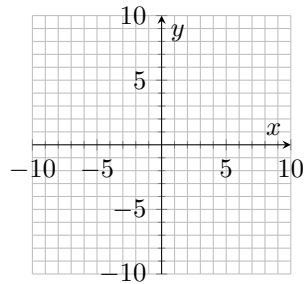
b. Find the domain and range of $f(x)$.



3. Let $g(x) = \log_4(x - 1)$.

a. Graph $g(x)$.

b. Find the domain and range of $g(x)$.



4. Use the properties of logarithms to expand the expression as much as possible.

a. $\log_7(3y)$

b. $\log\left(\frac{10^6}{x}\right)$

5. Use the properties of logarithms to rewrite the expression as a single logarithm.

a. $\log_c(16) - \log_c(2)$

b. $\ln(2) + 6\ln(x)$

6. Solve for x . Write your answer as an exact and approximate value rounded to three decimal places [4 pts each].

a. $4^x = 3$

b. $e^{2x+1} = 10$

c. $\log_6 x = 2$

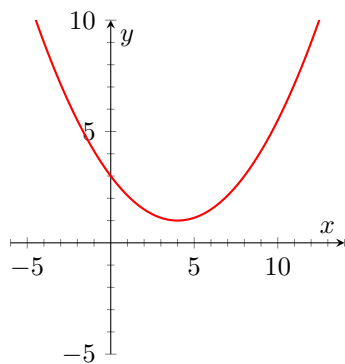
d. $\ln(40) = 3 - \ln(x)$

e. $\log_2(x+3) + \log_2(x-3) = 3$

7. If \$500 is invested in an account at 5% annual interest compounded monthly, what is the amount in the account after 9 years?

8. How long will it take \$500 to grow to \$1500 if it is invested at 10% compounded continuously?

9. Find and graph the focus, directrix, and vertex of the parabola $y = \frac{1}{8}(x - 4)^2 + 1$.



10. Find and graph the equation of the parabola with focus $(-4, -5)$ and directrix $y = -3$. ($p = -1$)

